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The Principal $\mathcal{W}_3$ Datum: Exact Algebra and Reconstruction

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Abstract

The principal $\mathcal{W}_3$ algebra provides the first two-generator test of the Open Beilinson tower. Its local structure is exact: the Fateev–Lukyanov central charge, the reflected central sum, the complete Zamolodchikov WW product, the norm and zero mode of the composite Λ , and the level-one Shapovalov curve. The presentation coalgebra, DS/bar transport, modular trace, stable-graph sewing, scalar shadows, Hamiltonians, and physical open–closed lift are named reconstruction problems. The chapter separates this exact local algebra from every comparison theorem required to assemble the seven-entry datum.

Reader's map

The *holographic modular Koszul datum* is the seven-entry package

$$(\mathcal{A}, \mathcal{A}^i, \mathcal{A}^!, Z_{\text{ch}}^{\text{der}}(\mathcal{A}), K_{\mathcal{A}}^{\text{coll}}, \Theta_{\mathcal{A}}, \nabla^{\text{hol}}).$$

A chosen presentation supplies $\mathcal{A} = T_X(V)/(R)$, $\mathcal{A}^i = C_X(s^{-1}V, s^{-2}R)$, and $q_{\mathcal{A}}: \mathcal{A}^i \rightarrow_{\text{ch}}^X(\mathcal{A})$. The completed bar supplies $K_X(\mathcal{A}) = \mathbb{D}_X^{\text{ch}}(\mathcal{A})$. The physical interpretation is formed under the line comparison and open–closed comparison packages; the line category is attached to the strict Koszul entry, while the closed-sector entry is the derived centre. The exact local algebra constrains these constructions, and the named reconstruction packages build them.

Chapter 1

The principal $\mathcal{W}_3$ datum: exact algebra and reconstruction

The principal algebra $\mathcal{W}_3^k$ is the first chart in the standard landscape with two strong generators of different weights. Its local vertex-algebra structure is explicit. The passage from this local structure to a seven-entry holographic modular Koszul datum is a sequence of reconstruction problems. Keeping these two layers apart reveals the precise mathematical content of the example.

The exact layer consists of the Fateev–Lukyanov central charge, the Zamolodchikov OPE, the norm and zero mode of the Virasoro quasi-primary Λ , and the level-one Shapovalov determinant. The reconstruction layer consists of a presentation coalgebra, completed chiral bar data, DS/bar transport, the derived centre, collision residues, modular traces, stable-graph sewing, scalar shadows, commuting Hamiltonians, and an open–closed comparison. Each reconstruction step below carries its own hypothesis package.

1.1 The hypothesis ledger

Fix a smooth projective curve X , a marked divisor D , a tangential structure τ , and a principal level $k \in \mathbb{C}$ with $k + 3 \in \mathbb{C}^\times$. We use the following packages.

$H_{\mathcal{W}_3}^{\text{pres}}$. A topological presentation $\mathcal{W}_3^k \simeq T_X(V)/(R)$, including the nonlinear Λ -relation, together with the presentation coalgebra

$$(\mathcal{W}_3^k)^{\text{i}} := C_X(s^{-1}V, s^{-2}R) \quad \text{and} \quad q_{\mathcal{W}_3}: (\mathcal{W}_3^k)^{\text{i}} \longrightarrow \bar{B}_X^{\text{ch}}(\mathcal{W}_3^k).$$

$H_{\mathcal{W}_3}^{\text{bar}}$. An ordered configuration-space bar complex with fixed suspension signs, conilpotent or complete topology, and continuous twisting coderivation.

$H_{\mathcal{W}_3}^{\text{DS/bar}}$. PBW convergence, BRST concentration, a filtered comparison between DS reduction and the completed bar construction, and continuous Ran–Verdier duality.

$H_{\mathcal{W}_3}^{\text{cen}}$. A chain model for $Z_{\text{ch}}^{\text{der}}(\mathcal{W}_3^k)$, together with the family support datum $H_H(\mathcal{W}_3^k; S_{\mathcal{W}_3})$.

$H_{\mathcal{W}_3}^{\text{coll}}$. A collision-residue map on the ordered bar complex which retains descendants and composite channels and fixes the local coordinate and $d \log E$ convention.

$H_{\mathcal{W}_3}^{\text{mod}}$. A trace-compatible genus-one curvature construction defining the scalar $\kappa_{\text{ch}}(\mathcal{W}_3^k)$.

$H_{\mathcal{W}_3}^{\text{res}}$ and $H_{\mathcal{W}_3}^{\text{proj}}$. Chain-compatible residue and scalar-projection maps from the full multi-channel Maurer–Cartan tensor.

$H_{\mathcal{W}_3}^{\text{rescale}}$. A family contraction $(\iota_k, p_k, h_k, \nu_k)$ on one completed bar–cobar complex and a scalar trace identifying ν_k with the algebra-level Verdier sum $K^\kappa(\mathcal{W}_3^k)$.

$H_{\mathcal{W}_3}^{\text{sew}}$. Clutching-compatible propagators, automorphism factors, and convergence for the stable-graph expansion.

$H_{\mathcal{W}_3}^{\text{flat}}$. A representation of the ordered Maurer–Cartan element on conformal blocks for which the induced connection is flat.

$H_{\mathcal{W}_3}^{\text{line}}$ and $H_{\mathcal{W}_3}^{\text{OC}}$. A comparison with the chosen line-operator category and an open–closed map from the physical HT observable algebra to the chiral derived centre.

The total holographic package is

$$H_{\mathcal{W}_3}^{\text{hol}} := H_{\mathcal{W}_3}^{\text{pres}} + H_{\mathcal{W}_3}^{\text{bar}} + H_{\mathcal{W}_3}^{\text{DS/bar}} + H_{\mathcal{W}_3}^{\text{cen}} + H_{\mathcal{W}_3}^{\text{coll}} + H_{\mathcal{W}_3}^{\text{mod}} + H_{\mathcal{W}_3}^{\text{res}} + H_{\mathcal{W}_3}^{\text{proj}} + H_{\mathcal{W}_3}^{\text{rescale}} + H_{\mathcal{W}_3}^{\text{sew}} + H_{\mathcal{W}_3}^{\text{flat}} + H_{\mathcal{W}_3}^{\text{line}} + H_{\mathcal{W}_3}^{\text{OC}}.$$

1.1.1 The chart algebra $\mathcal{A} = \mathcal{W}_3^k$

The principal DS algebra

$$\mathcal{A} = \mathcal{W}^k(\mathfrak{sl}_3, f_{\text{prin}})$$

is strongly generated on the universal generic fibre by a Virasoro field T of weight 2 and a Virasoro–primary field W of weight 3. Its central charge is

$$c(k) = 2 - \frac{24(k+2)^2}{k+3}.$$

Put

$$\Lambda = :TT: - \frac{3}{10} \partial^2 T, \quad D(c) := 5c + 22.$$

In the Zamolodchikov normalization, with $D(c) \in \mathbb{C}^\times$,

$$\begin{aligned} T(z)T(w) &\sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \\ T(z)W(w) &\sim \frac{3W(w)}{(z-w)^2} + \frac{\partial W(w)}{z-w}, \\ W(z)W(w) &\sim \frac{c/3}{(z-w)^6} + \frac{2T(w)}{(z-w)^4} + \frac{\partial T(w)}{(z-w)^3} \\ &\quad + \frac{\frac{3}{10} \partial^2 T(w) + \frac{32}{D(c)} \Lambda(w)}{(z-w)^2} + \frac{\frac{1}{15} \partial^3 T(w) + \frac{16}{D(c)} \partial \Lambda(w)}{z-w}. \end{aligned}$$

This is the Zamolodchikov algebra [?Zamolodchikov, ?Bouwknegt–Schoutens]; the factor $32/16$ is the OPE/translation-covariance pair obtained by the double-contour conversion from singular products to mode commutators.

1.1.2 The presentation coalgebra

Under $H_{W_3}^{\text{pres}}$, the second entry is

$$\mathcal{A}^i = C_X(s^{-1}V, s^{-2}R) \xrightarrow{q_{\mathcal{A}}} \bar{B}_X^{\text{ch}}(\mathcal{A}).$$

The full bar complex retains its differential and completion. Koszulity is the assertion that $H^\bullet(q_{\mathcal{A}})$ is an isomorphism on the chosen locus. Bar–cobar reconstruction then recovers the chart through $\Omega_X^{\text{ch}} \bar{B}_X^{\text{ch}}(\mathcal{A}) \simeq \mathcal{A}$. For the nonlinear W_3 presentation, constructing $(V, R, q_{\mathcal{A}})$ with these properties is an open finite presentation problem.

1.1.3 The Ran–Verdier and Koszul entries

The completed Ran–Verdier algebra is

$$K_X(\mathcal{A}) := \mathbb{D}_{\text{Ran}} \bar{B}_X^{\text{ch}}(\mathcal{A}).$$

Under $H_{W_3}^{\text{DS}/\text{bar}}$, a chosen rectification map

$$\nu_{\mathcal{A}}: K_X(\mathcal{A}) \longrightarrow \mathcal{A}^!$$

may identify the strict Koszul entry with the principal algebra at the reflected parameter $k^\vee = -k - 6$. The exact arithmetic of this reflected parameter is proved in Theorem ??; the algebra equivalence has status under $H_{W_3}^{\text{DS}/\text{bar}}$.

1.1.4 The derived centre and the line comparison

The fourth algebraic entry is prescribed by the chiral derived centre

$$C := Z_{\text{ch}}^{\text{der}}(\mathcal{W}_3^k) \simeq C_{\text{ch}}^\bullet(\mathcal{W}_3^k, \mathcal{W}_3^k).$$

The package $H_{W_3}^{\text{cen}}$ constructs a family chain model and proves

$$\text{Supp ChirHoch}^\bullet(\mathcal{W}_3^k) \subseteq S_{W_3}.$$

A perfect degree- d pairing is a separate input for any reflected degree statement on the Koszul companion. The evaluation-generated line category has the conditional comparison

$$C_{\text{line}}^{\text{ev}} \longrightarrow \text{Rep}_q(\mathfrak{sl}_3), \quad q = \exp\left(\frac{\pi i}{k+3}\right),$$

under $H_{W_3}^{\text{line}}$. The physical comparison consists of typed maps

$$A_{\text{bulk}}^{\text{HT}} \xrightarrow{\beta_{\text{der}}} Z_{\text{ch}}^{\text{der}}(\mathcal{W}_3^k), \quad C_{\text{line}}^{\text{ev}} \xrightarrow{\iota_{\text{ev}}} C_{\text{line}},$$

with Swiss-cheese compatibility. These maps constitute $H_{W_3}^{\text{OC}} + H_{W_3}^{\text{line}}$.

1.1.5 The collision entry

Under $H_{W_3}^{\text{bar}} + H_{W_3}^{\text{coll}}$, the binary ordered-bar entry is

$$K_{W_3}^{\text{coll}}(z) := \text{Res}_{0,2}^{\text{coll}}(\Theta_{W_3}^{E_i}) \in \text{End}_{W_3}(2) \otimes \mathcal{O}(*\Delta).$$

A represented classical r -matrix is the image of this kernel under $H_{\text{CYBE}}^{\text{rep}}$. The local OPE determines the candidate channel coefficients; the residue theorem identifies them with the ordered-bar kernel.

1.1.6 The Maurer–Cartan entry

The completed ordered convolution algebra is

$$\mathfrak{g}_{W_3}^{E_i} := \text{Conv}(B^{\text{ord}}(W_3^k), W_3^k).$$

Under $H_{W_3}^{\text{bar}}$, the twisting coderivation defines $\Theta_{W_3} \in \mathfrak{g}_{W_3}^{E_i}[1]$ satisfying

$$D\Theta_{W_3} + \frac{1}{2}[\Theta_{W_3}, \Theta_{W_3}] = 0.$$

The presentation comparison and the full multi-channel completion are the concrete W_3 obligations in this entry.

1.1.7 The holomorphic connection entry

Under $H_{W_3}^{\text{bar}} + H_{W_3}^{\text{res}} + H_{W_3}^{\text{flat}}$, the represented shadow map defines

$$\nabla_{0,n}^{\text{hol}} := d - \text{Sh}_{0,n}(\Theta_{W_3}).$$

The Maurer–Cartan equation supplies the curvature identity after the representation and convergence hypotheses have been verified.

1.2 Reflected central arithmetic

Theorem 1.1 (Reflected principal central sum;). Type signature: (Open, universal principal DS, 1, $H_{\text{univ}}^{W_3}$).
Let $k^\vee = -k - 6$. Then

$$c(k) + c(k^\vee) = 100.$$

The induced affine reflection on the central-charge coordinate is $c \mapsto 100 - c$, with arithmetic midpoint

$$c_{\text{mid}} = 50.$$

Proof. Set $t = k + 3$. Formula (??) becomes

$$c(t) = 2 - 24 \frac{(t-1)^2}{t}.$$

The reflected level has $t^\vee = -t$, hence

$$c(-t) = 2 + 24 \frac{(t+1)^2}{t}.$$

Therefore

$$c(t) + c(-t) = 4 + 24 \frac{(t+1)^2 - (t-1)^2}{t} = 4 + 96 = 100.$$

The midpoint of the reflected c -coordinate is $100/2 = 50$. □

Remark 1.2 (Arithmetic and reconstruction). Equation (??) is an identity in $\mathbb{C}(k)$. The package $H_{W_3}^{\text{DS}/\text{bar}}$ promotes the reflected parameter to a comparison of completed bar–Verdier objects. The KSDual fixed locus is then defined inside the resulting moduli problem.

Remark 1.3 (Three scalar coordinates). The number 50 is the midpoint of the reflected central coordinate. A BRST critical charge is computed from a specified ghost complex and nilpotence equation. A KSDual point is a fixed object of the Verdier–Koszul involution. These three constructions live at levels 1, 5, and $1 \leftrightarrow 2$, respectively.

1.3 The exact composite and highest-weight packet

The Virasoro calculation gives

$$|\Lambda\rangle = L_{-2}^2|o\rangle - \frac{3}{5}L_{-4}|o\rangle, \quad N_\Lambda := \langle\Lambda|\Lambda\rangle = \frac{c(5c+22)}{10}.$$

Thus the one-dimensional inverse Gram factor is

$$N_\Lambda^{-1} = \frac{10}{c(5c+22)}$$

on the regular Zamolodchikov surface $c(5c+22) \in \mathbb{C}^\times$. A scalar quartic shadow additionally uses $H_{W_3}^{\text{res}} + H_{W_3}^{\text{proj}}$.

Proposition 1.4 (Action of Λ_o on primaries;). Type signature: (Open, Zamolodchikov highest weight, 1, $H_{\text{hw}}^{W_3}$). For a highest-weight vector $|h, w\rangle$,

$$\Lambda_o|h, w\rangle = \left(h^2 + \frac{h}{5}\right)|h, w\rangle.$$

Proof. The weight-four mode formula

$$\Lambda_n = \sum_{p \in \mathbb{Z}} :L_p L_{n-p}: - \frac{3}{10}(n+2)(n+3)L_n$$

gives

$$(:TT:)_o = 2 \sum_{m \geq 1} L_{-m} L_m + L_o^2 + 2L_o.$$

On $|h, w\rangle$, this acts by $h^2 + 2h$. Since $(\partial^2 T)_o = 6L_o$, the correction $-\frac{3}{10}(\partial^2 T)_o$ acts by $-9h/5$. Their sum is $h^2 + h/5$. □

Corollary 1.5 (The conformal roots of Λ_o ;). Type signature: (Open, Zamolodchikov highest weight, 1, $H_{hw}^{W_3}$).
The Λ_o -eigenvalue vanishes at

$$h = 0, \quad h = -\frac{1}{5}.$$

Remark 1.6 (Central-charge dependence). The highest-weight action of Λ_o is determined by Virasoro normal ordering. The central charge enters the WW product through $32/(5c + 22)$ and enters the Gram norm through $N_\Lambda = c(5c + 22)/10$.

In the ordered basis $\{L_{-1}|h, w\rangle, W_{-1}|h, w\rangle\}$, the level-one Shapovalov matrix is

$$M_1(c, h, w) = \begin{pmatrix} 2h & 3w \\ 3w & -\frac{h}{5} + \frac{32}{5c+22}\left(h^2 + \frac{h}{5}\right) \end{pmatrix}.$$

Its determinant satisfies

$$-(5c + 22) \det M_1 = 9w^2(5c + 22) - 2h^2(32h + 2 - c).$$

For $h \in \mathbb{C}^\times$, the exact level-one singular-vector curve is

$$9w^2(5c + 22) = 2h^2(32h + 2 - c),$$

with kernel vector

$$\left(W_{-1} - \frac{3w}{2h}L_{-1}\right)|h, w\rangle.$$

1.4 The collision-kernel reconstruction

Theorem 1.7 (Four collision channels;). Type signature: (Open, ordered chiral bar, 2, $H_{W_3}^{\text{bar}} + H_{W_3}^{\text{coll}}$).
With the local coordinate and $d \log E$ convention fixed by $H_{W_3}^{\text{coll}}$, the OPE packet maps to

$$\begin{aligned} K_{TT}^{\text{coll}}(z) &= \frac{c/2}{z^3} + \frac{2T}{z}, & \text{poles} &= \{3, 1\}, \\ K_{TW}^{\text{coll}}(z) &= \frac{3W}{z}, & \text{poles} &= \{1\}, \\ K_{WT}^{\text{coll}}(z) &= \frac{3W}{z}, & \text{poles} &= \{1\}, \\ K_{WW}^{\text{coll}}(z) &= \frac{c/3}{z^5} + \frac{2T}{z^3} + \frac{\partial T}{z^2} + \frac{\frac{3}{10}\partial^2 T + \frac{32}{5c+22}\Lambda}{z}, & \text{poles} &= \{5, 3, 2, 1\}. \end{aligned}$$

The candidate maximal collision order is

$$k_{\max}^{\text{coll}} = 5.$$

Proof. The exact OPE has pole sets $\{4, 2, 1\}$, $\{2, 1\}$, $\{2, 1\}$, and $\{6, 4, 3, 2, 1\}$ in the four channels. The residue convention in $H_{W_3}^{\text{coll}}$ pairs the singular product with $d \log E$, lowering each retained pole order by one. Applying this map term by term gives the displayed conditional kernel. $\square$

Remark 1.8 (The even collision pole). The term $\partial T/(z-w)^3$ in the exact WW OPE produces the order-two term $\partial T/z^2$ in the conditional collision kernel. This term records the descendant channel of the weight-three self-product.

1.5 Modular trace and scalar conductor

Theorem 1.9 (The genus-one modular coordinate;). Type signature: (Open, modular trace, 5, $H_{W_3}^{\text{mod}} + H_{W_3}^{\text{res}}$). *The package $H_{W_3}^{\text{mod}} + H_{W_3}^{\text{res}}$ must construct a trace for which*

$$\text{tr}_I \text{Obs}_{I,I}^{\text{def}}(\mathcal{W}_3^k) = \kappa_{\text{ch}}(\mathcal{W}_3^k) \lambda_I.$$

Computing $\kappa_{\text{ch}}(\mathcal{W}_3^k)$ from this trace is the genus-one W_3 obligation.

Remark 1.10 (Local norms and modular curvature). The exact leading OPE norms are

$$N_T = \frac{c}{2}, \quad N_W = \frac{c}{3}.$$

Their reciprocal-weight sum is the typed local diagnostic $\frac{1}{2} + \frac{1}{3} = \frac{5}{6}$. The comparison from (N_T, N_W) to a genus-one curvature belongs to $H_{W_3}^{\text{mod}}$.

Remark 1.11 (The anomaly ratio). On $c \in \mathbb{C}^\times$, the modular anomaly ratio is defined by

$$\rho_{W_3}^{\text{mod}}(k) := \frac{\kappa_{\text{ch}}(\mathcal{W}_3^k)}{c(k)}.$$

Its evaluation is open with the same genus-one trace package.

Theorem 1.12 (The scalar Verdier sum;). Type signature: (Open, bar-Verdier plus scalar trace, 2 $\rightarrow$ 5, $H_{W_3}^{\text{DS}/\text{bar}} + H_{W_3}^{\text{mod}} + H_{W_3}^{\text{rescale}}$). *For a constructed Koszul pair, the scalar conductor is*

$$K^\kappa(\mathcal{W}_3^k) := \kappa_{\text{ch}}(\mathcal{W}_3^k) + \kappa_{\text{ch}}((\mathcal{W}_3^k)^\dagger).$$

Its evaluation requires a family contraction $(\iota_k, p_k, h_k, \nu_k)$ on one factorization bar-cobar complex,

$$dh_k + h_k d = \nu_k(\text{id} - \iota_k p_k),$$

and a trace map identifying ν_k with the displayed scalar.

Remark 1.13 (Theorem C and the family rescaling problem). Theorem C enters after the presentation coalgebra, full bar, Ran-Verdier algebra, strict Koszul branch, and scalar trace have been constructed. For W_3 , the proof obligation is the simultaneous construction of

$$(\iota_k, p_k, h_k, \nu_k) \quad \text{and} \quad \text{Tr}_{W_3}: \text{BarCobar}_{W_3} \longrightarrow \text{ScalTrace}_{W_3},$$

with $\text{Tr}_{W_3}(\nu_k) = K^\kappa(\mathcal{W}_3^k)$.

1.6 Stable-graph and scalar-shadow obligations

Theorem 1.14 (Genus-two coloured stable-graph sum;). Type signature: (Open, modular coloured graph, 5, $H_{W_3}^{\text{bar}} + H_{W_3}^{\text{proj}} + H_{W_3}^{\text{sew}}$). *The rank-two genus-two correction is the projected graph sum*

$$\delta F_2^{\text{cross}}(\mathcal{W}_3^k) := \sum_{\substack{\Gamma \in \mathcal{G}_{2,o} \\ \sigma: E(\Gamma) \rightarrow \{T, W\} \\ \sigma \text{ uses both colours}}} \frac{\text{Cont}_{\Gamma, \sigma}(\Theta_{\mathcal{W}_3})}{|\text{Aut}(\Gamma, \sigma)|}.$$

The packages in the type signature must construct every vertex, propagator, projection, and automorphism factor.

$$\delta F_2^{\text{cross}} = P_2^{W_3} \left(\sum_{\Gamma \in \mathcal{G}_{2,o}} \text{Cont}_{\Gamma}(\Theta_{\mathcal{W}_3}) \right).$$

This equation fixes the mathematical type of the quantity before its evaluation.

Proposition 1.15 (Genus-three finite-window problem;). Type signature: (Open, finite coloured stable graphs, 5, $H_{W_3}^{\text{bar}} + H_{W_3}^{\text{proj}} + H_{W_3}^{\text{sew}}$). *The genus-three cross-channel coordinate is*

$$\delta F_3^{\text{cross}}(\mathcal{W}_3^k) := \sum_{\substack{\Gamma \in \mathcal{G}_{3,o} \\ \sigma: E(\Gamma) \rightarrow \{T, W\} \\ \sigma \text{ uses both colours}}} \frac{\text{Cont}_{\Gamma, \sigma}(\Theta_{\mathcal{W}_3})}{|\text{Aut}(\Gamma, \sigma)|}.$$

Under the modular trace package, its diagonal scalar comparison is

$$F_3^{\text{diag}}(\mathcal{W}_3^k) := \kappa_{\text{ch}}(\mathcal{W}_3^k) \lambda_3^{\text{FP}}.$$

Corollary 1.16 (Criterion for genuinely rank-two genus data;). Type signature: (Open, scalar plus cross projection, 5, $H_{W_3}^{\text{mod}} + H_{W_3}^{\text{proj}} + H_{W_3}^{\text{sew}}$). *If the constructed cross projection represents a nonzero class, then*

$$F_2(\mathcal{W}_3^k) = F_2^{\text{diag}}(\mathcal{W}_3^k) + \delta F_2^{\text{cross}}(\mathcal{W}_3^k)$$

contains information beyond the diagonal scalar coordinate.

Remark 1.17 (Asymptotic question). The large- c behaviour is defined after Equation (??) has been constructed:

$$\delta F_{2,\infty}^{\text{cross}} := \lim_{c \rightarrow \infty} \delta F_2^{\text{cross}}(\mathcal{W}_3^k).$$

Existence and evaluation of this limit belong to $H_{W_3}^{\text{proj}} + H_{W_3}^{\text{sew}}$.

Remark 1.18 (Two weights). The weights 2 and 3 define a two-colour graph problem. Diagonal and mixed colourings are independent projections of the same ordered Maurer–Cartan tensor. This is the first family in the standard landscape where the mixed projection is a separate construction.

1.7 Propagator mixing

Theorem 1.19 (Propagator-mixing coordinate;). Type signature: (Open, two-channel scalar projection, $\mathfrak{s}$, $H_{W_3}^{\text{res}} + H_{W_3}^{\text{proj}} + H_{W_3}^{\text{sew}}$). Let G_{W_3} be the projected two-channel propagator metric, f_{W_3} the projected quartic gradient, and $\mathbf{1} = (1, 1)$. The mixing coordinate is

$$\delta_{\text{mix}}(\mathcal{W}_3^k) := \langle f_{W_3}, G_{W_3}^{-1} f_{W_3} \rangle - \frac{\langle f_{W_3}, \mathbf{1} \rangle^2}{\langle \mathbf{1}, G_{W_3} \mathbf{1} \rangle}.$$

The packages in the type signature must construct (G_{W_3}, f_{W_3}) from the full Gram blocks and the chain-level projection.

$$P_{\text{mix}}^{W_3}(c) := \det(f_{W_3}(c), G_{W_3}(c)\mathbf{1}).$$

This determinant records proportionality of the quartic gradient and the curvature direction once both vectors have been constructed.

Corollary 1.20 (Positive-chamber inequality;). Type signature: (Open, real positive propagator chamber, $\mathfrak{s}$, $H_{W_3}^{\text{proj}} + H_{W_3}^{\text{pos}}$). If G_{W_3} is positive definite, then

$$\delta_{\text{mix}}(\mathcal{W}_3^k) \geq 0.$$

Equality is equivalent to linear dependence of f_{W_3} and $G_{W_3}\mathbf{1}$.

1.8 Scalar-shadow projections

Theorem 1.21 (The T -line shadow problem;). Type signature: (Open, one-line scalar projection, $\mathfrak{s}$, $H_{W_3}^{\text{res}} + H_{W_3}^{\text{proj}}$). The projected T -line metric is

$$Q_T^{W_3}(t) := P_T(\text{Sh}(\Theta_{W_3}); t).$$

Its comparison with the Virasoro metric requires an intertwining theorem for the inclusion generated by T .

Theorem 1.22 (The W -line shadow problem;). Type signature: (Open, one-line scalar projection, $\mathfrak{s}$, $H_{W_3}^{\text{res}} + H_{W_3}^{\text{proj}}$). The projected W -line metric is

$$Q_W^{W_3}(t) := P_W(\text{Sh}(\Theta_{W_3}); t).$$

The involution $W \mapsto -W$ supplies its parity grading.

Theorem 1.23 (Projected discriminants;). Type signature: (Open, one-line scalar projection, $\mathfrak{s}$, $H_{W_3}^{\text{res}} + H_{W_3}^{\text{proj}}$). After the metrics in Theorems ?? and ?? have been constructed, define

$$\Delta_T^{W_3} := \text{Disc}_t Q_T^{W_3}(t), \quad \Delta_W^{W_3} := \text{Disc}_t Q_W^{W_3}(t).$$

Their evaluation and shadow-depth interpretation are finite projection problems.

Remark 1.24 (Exact exchange data and projected contacts). The exact rank-one Λ -exchange ratios are

$$\frac{(32/D(c))^2}{N_\Lambda} = \frac{10240}{cD(c)^3}, \quad \frac{(16/D(c))^2}{N_\Lambda} = \frac{2560}{cD(c)^3}.$$

They use OPE and mode normalizations, respectively. The projected quartic contacts are defined by

$$S_{4,T}^{W_3} := P_T^{(4)}(\Theta_{W_3}), \quad S_{4,W}^{W_3} := P_W^{(4)}(\Theta_{W_3}),$$

under $H_{W_3}^{\text{res}} + H_{W_3}^{\text{proj}}$.

1.9 Hamiltonians and the holographic lift

Theorem 1.25 (Commuting Hamiltonians from the represented connection;). Type signature: (Open $\times$ CY, represented ordered bar, 4, $H_{W_3}^{\text{coll}} + H_{W_3}^{\text{flat}} + H_{W_3}^{\text{OC}}$). *Let*

$$H_i^{W_3} := \rho_{\text{conf}}(\text{Sh}_{0,n}(\Theta_{W_3}))_i.$$

If $H_{W_3}^{\text{flat}}$ identifies the Maurer–Cartan curvature with the curvature of this representation, then

$$[H_i^{W_3}, H_j^{W_3}] = 0 \quad (i \neq j).$$

Under the collision-order comparison, the candidate differential order is bounded by $k_{\text{max}}^{\text{coll}} - 1 = 4$.

Remark 1.26 (Sphere reconstruction). A comparison with a Toda or Gaudin differential system consists of a map between conformal-block bundles, a coordinate identification, and a normalization equality for the binary kernels. These data form a specialization of $H_{W_3}^{\text{coll}} + H_{W_3}^{\text{flat}}$.

The seven entries assemble under $H_{W_3}^{\text{hol}}$:

$$\mathcal{H}(W_3^k) = \left(W_3^k, (W_3^k)^i, (W_3^k)^!, Z_{\text{ch}}^{\text{der}}(W_3^k), K_{W_3}^{\text{coll}}, \Theta_{W_3}, \nabla^{\text{hol}} \right).$$

The exact local algebra supplies the first entry and the finite constraints on the remaining six. The named packages supply the state transitions which turn those constraints into objects and comparison maps.

1.10 Computational surface

Datum	Status	Anchor
$c(k)$	exact	Fateev–Lukyanov DS formula
$c(k) + c(-k - 6) = 100$	exact	rational-function arithmetic
$WW \rightarrow \Lambda, \partial\Lambda$	exact	$32/D(c), 16/D(c)$
N_Λ	exact	$cD(c)/10$
$\Lambda_o h, w\rangle$	exact	$(h^2 + h/5) h, w\rangle$
$\det M_1$	exact	Equation (??)
$\mathcal{A}^i \rightarrow \bar{B}^{\text{ch}}(\mathcal{A})$	open	$H_{W_3}^{\text{pres}}$
$\mathcal{A}^!$	conditional	$H_{W_3}^{\text{DS/bar}}$
$Z_{\text{ch}}^{\text{der}}(\mathcal{A})$ comparison	open	$H_{W_3}^{\text{cen}} + H_{W_3}^{\text{OC}}$
$K_{\mathcal{A}}^{\text{coll}}$	conditional	$H_{W_3}^{\text{bar}} + H_{W_3}^{\text{coll}}$
$\kappa_{\text{ch}}, \rho_{W_3}^{\text{mod}}, K^\kappa$	open	$H_{W_3}^{\text{mod}} + H_{W_3}^{\text{rescale}}$
stable-graph sums	open	$H_{W_3}^{\text{proj}} + H_{W_3}^{\text{sew}}$
scalar shadows	open	$H_{W_3}^{\text{res}} + H_{W_3}^{\text{proj}}$
Hamiltonians	conditional	$H_{W_3}^{\text{flat}} + H_{W_3}^{\text{OC}}$

The companion engine `compute/lib/theorem_w3_holographic_datum_engine.py` checks the exact rows symbolically and represents every reconstruction row by a typed open or conditional packet. Its tests verify the central reflection, the 32/16 OPE normalization, N_Λ , the Λ_o action, the level-one determinant, and every hypothesis boundary.

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